Kanthan (Fe-er-Al) –Heating element in furnaces.

Carbon – Brushes in electrical machines.

9.5.2 Comparison of Copper and Aluminium

FactorS	Aluminium	Copper
Colour	Silverly white	Reddish brown
Density	2700 kg/m^3	8900° kg/m^3
Melting point	660°C	1085°C
Resistivity	2.65 x10^{-6}Ω-m	1.72 x10^{-6}Ω-m

9.5.3 Advantage of Aluminium

- Cheaper - Low Cost
- Lighter - 1/3 weight of copper.
- Softer.
- Non-Reactive to acids.
- Availability - Aluminium is third most abundant element in nature. 7.28% of earth crust is aluminium.

9.5.4 Disadvantage of Aluminium

- Higher Resistivity : Hence aluminium is not used for winding. If used, the wire must be of large size to reduce I^2R losses and machine size also increases.
- High contact Resistance.
- Poor Tensile Strength - Aluminium cannot be used directly as overhead transmission line as it cannot be stretched.
- Possibilities for Loose contacts - Aluminium cannot be soldered by conventional method.

9.6 INSULATING MATERIALS

9.6.1 Solid Insulating Materials

A) Plastics

- It is an organic polymer, which can be moulded into and desired shape and size with the help of heat, pressure or both.

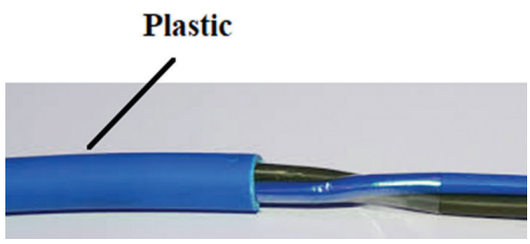


Fig 9.6 *Application of Plastic*

- The plastic in liquid forms is known as resin.
- There are two types of plastics namely Thermo plastic and Thermo set plastic .
- Thermo plastic can be softened and hardened by heating and cooling by any number of times. (e.g) Polythene, PVC, Acrylic.
- Thermo set plastic cannot be softened once they are moulded. It is formed by condensation and polymerization. Eg. Polyester, Bakelite, Epoxy

b) Ceramics

- Ceramics are inorganic, non-metallic, covalent compounds.
- They are produced from earthy material (clay) by the action of fire
- Clay product, refractories and glasses are types of ceramics.

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